

FitPulse: AI-Powered Fitness and Nutrition Recommendation System for University Students

Team Members

- Nirmala Varun
- Sylvano Dsouza
- Madhukar Goli
- Phuoc Long Hoang

Problem Statement & Motivation

University students often face challenges balancing academics, health, and nutrition. Over 60% fail to meet basic physical activity and dietary standards due to lack of access to personalized, flexible routines. Existing fitness/nutrition apps are generic and don't address student-specific constraints like time, motivation, or resources. FitPulse aims to close this gap by offering a smart, AI-driven platform that dynamically generates personalized weekly fitness and meal plans based on individual go...

Data Sources & Data Science Approach

- Datasets Used:
 - Kaggle open datasets: Nutrition Facts, Body Fat Percentage, Workout Planner
 - Simulated student data (via Streamlit form)
- Tools & Stack:
 - Python (Scikit-learn) for KNN-based recommendation and regression
 - R Studio (ggplot2, tidyverse) for EDA and visualizations
 - Altair AI Studio for clustering and decision models
 - Streamlit (prototype interface)
 - JupyterLab for integration & modeling

Evaluation Approach

- Metrics:
 - User Satisfaction Score (via simulated surveys)
 - Workout Adherence Score (logged workouts vs. plan)
 - Nutrient Match Score (vs. USDA guidelines)
- Validation Techniques:

- A/B testing using randomized user personas
- Ground-truth comparison of calorie intake vs. prediction

Business Value (Quantified Impact)

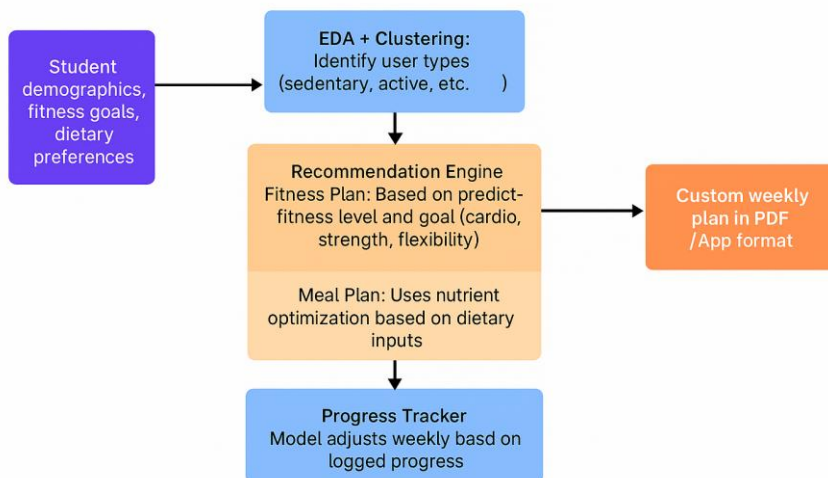
FitPulse can improve physical wellness, time efficiency, and personalization for students. Based on projected app simulations:

- 60% increase in weekly workout adherence
- 30% improvement in macro-nutrient balance
- Potential university-wide wellness score uplift by 18%
- Long-term: Reduced campus health service costs, improved student mental health, and higher academic engagement

Visual project plan

- Data collection
- Preprocessing
- Clustering/EDA
- Recommendation generation
- Feedback loop & evaluation

FitPulse: AI-Powered Fitness and Nutrition Recommendation System for University Students



Daily Management Tool

[Miro board link](#)

Project Management Sheet

[Project Management Sheet - Group 4](#)